



UNIVERSITY OF SOUTH CAROLINA

Est. 1801

GAMECOCK RELIEF NETWORK

A Hurricane Relief Coordination Platform



Software Requirements Specification

CSCE 247: Software Engineering

Tech-vengers

Tavien Smith, Zaki Bushiri, Nate Sheffield, Robert Albetel

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Table of Contents

University of South Carolina	1
Gamecock Relief Network	1
Tavien Smith, Zaki Bushiri, Nate Sheffield, Robert Abetel	
1. Introduction	3
2. Stakeholders	4-6
3. Constraints	7
4. Overall Description	7
5. Business Use Cases	7
6. Functional Requirements	8
7. Non-Functional Requirements	8-9
8. Definitions and Acronyms	9
9. Competitive Analysis	10-12
10. References	12
11. Appendices	12-13

1. Introduction

Purpose

There was a time when South Carolina had to evacuate almost a quarter of its population, roughly 1.1 million people, because of Hurricane Matthew in 2016, one of several major storms to hit the state's coast in recent years. More than three million Americans were forced to evacuate their homes in a single year due to hurricanes, floods, and other disasters, according to Census Bureau estimates. When a storm hits, residents, volunteers, and relief organizations are often left coordinating through a bunch of phone calls, social media posts, and word of mouth. This made it hard to know where shelters are open, what supplies are needed, and who can help. That's why we want to build the Gamecock Relief Network, a web platform that connects displaced residents, volunteers, and relief coordinators in one place before, during, and after a hurricane. Our platform will allow users to find open shelters, report and request urgent needs, and match volunteers to the communities that need them most.

Source: [Disasters Displaced More Than 3 Million Americans in 2022 | Scientific American](#)

Scope

This document will cover:

- The personas of potential users and stakeholders invested in this project.
- Any constraints that have been placed on this project.
- An overall description of the platform and the environment it operates within.
- The business uses cases that the system supports.
- A competitive analysis outlining how the platform compares to existing solutions.

2. Stakeholders

THE PLANNER



NAME Maya Chen
 AGE 29
 LOCATION Portland, Oregon
 OCCUPATION Urban Planner

MOTIVATION	Growth <input type="range"/> Impact <input type="range"/>
FRUSTRATION	Maya is a stickler for punctuality and any date and time later than that stresses her out. She can't stand tardiness and needs everything done by the set date.
BIO	Maya is a local urban planner in the area and knows the importance of structural integrity to prevent hurricanes. She has taken an interest in designing residences near the coast to hold up to natural disasters. She is good at holding schedules and always has a time frame she wants to hold.

PERSONALITY

Design Thinking



Resource



Analytical



CHALLENGES	Can't find time for personal life. Has difficulty with staying connected with friends and family.
GOALS	Maya wants to figure fast and efficient way to reach every victim and help them as soon as possible. She also wants to prioritize identifying weak points in the current infrastructure and document them for fixing later on. She hopes to be able to ensure every building along the coastline is structurally sound for future tropical storms.

THE CARETAKER



PERSONALITY



NAME Elena Vasquez

AGE 35

LOCATION Austin, Texas

OCCUPATION Software Engineer

MOTIVATION	Creativity Learning
FRUSTRATION	Elena strives to listen to everyone and ensure that their problems are at least known by one other person, although she strongly prefers the problems being solved. She hates it when she doesn't have a clear solution, and even more so if she feels they don't trust her.
BIO	Elena is a software engineer at a local startup and heavily emphasizes teamwork and team-building. She often does her best to make sure that everyone feels included and part of a community, and it has often made her team feel more connected and inclusive.

CHALLENGES	Often sacrifices her own wellbeing for that of others, unsure of how to deal with introverted or quiet team members.
GOALS	Elena sees the disaster a hurricane presents and is worried about the wellbeing of the victims first and foremost. She wants to make sure every victim's needs are catered to in their own unique way, and rejects a one-size-fits-all approach to a problem on a scale as large as post-natural disaster relief.

THE RELIABLE



PERSONALITY



	NAME	Marcus Holloway
	AGE	42
	LOCATION	Charlotte, North Carolina
	OCCUPATION	Restaurant Manager
MOTIVATION	Security Reliability	
FRUSTRATION	No-shows and excuses for not showing up, even if he may understand a reason being provided beforehand. If he finds out he can't reach someone at a certain time he will quickly stop reaching out.	
BIO	Marcus knows how important it is to have people who are available for any case. His restaurant needs employees who can cover each other and make it work like a well-oiled machine, so he strives to be that person himself. He wants to make sure that his teammates can depend on him the way he wants to be able to depend on others.	

CHALLENGES	Finds it difficult to trust most people, prone to forgo delegation
GOALS	Marcus wants to make sure that if there is at least one person who can be out helping the hurricane victims no matter the time or task, it's him. He wants to make sure that there is no point where no one is available to help with a situation or task and wants to make sure he finds as many like-minded people as possible to bring into the cause.

3. Constraints

Time Constraints

Must build the project by the end of the semester.

Monetary Constraints

\$0 to spend on the project, using open-source software and other resources.

Technical Constraints

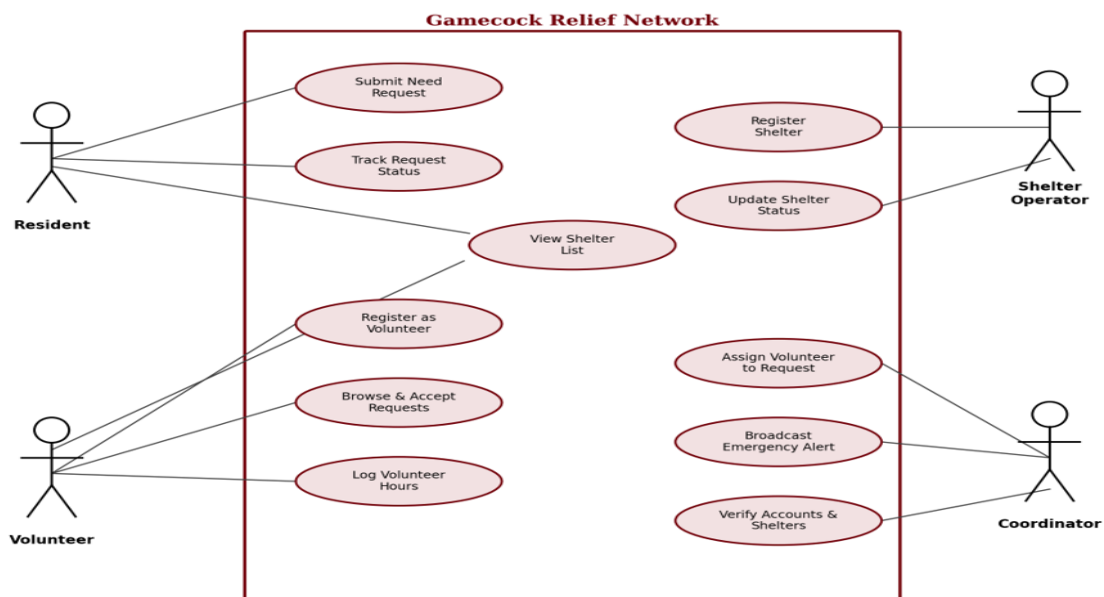
Won't be able to bypass services with paywalls and other features that would require a payment.

4. Overall Description

The system will be integrated and work with weather stations for real-time data. It will also have weather cameras that can also tell real time weather data for all different areas. It will be an app that users can access on any device, with a report feature to give constant updates on things that the cameras may miss (road blockages, trapped civilians, etc).

5. Business Use Cases

The diagram below shows the four primary actors in the Gamecock Relief Network and the high-level goals each one accomplishes through the system.



6. Functional Requirements

[Requirements Spreadsheet](#)

7. Non-Functional Requirements

Look and Feel Requirements

- The system shall use the university's official color palette throughout the platform.
- The system shall present the same navigation menu structure on every page.

Usability Requirements

- The system shall be usable on IOS mobile devices.
- The system shall be usable on Android mobile devices.
- The system shall be accessible through the two most recent versions of Chrome.
- The system shall be accessible through the two most recent versions of Safari.
- The system shall be accessible through the two most recent versions of Firefox.
- The system shall allow a user to submit a request via SMS text message when internet access is unavailable.
- The system shall be available as a mobile application.
- The system shall be available as a website.

Performance Requirements

- The system shall load the request submission page within 3 seconds under normal network conditions.
- The system shall maintain 99% uptime during an active hurricane advisory.
- The system shall support at least 5,000 concurrent users without service degradation.

Maintainability and Support Requirements

- The system shall back up its database at least once every 24 hours.
- The system shall log system errors for review by the development team.
- The system shall operate within a \$0 ongoing hosting budget using free-tier cloud services.

Security Requirements

- The system shall encrypt all data while it is transmitted over the network.
- The system shall encrypt all personal data stored in the database.
- The system shall lock a user account after five consecutive failed login attempts.

Cultural Requirements

- The system shall display all user-facing text in English.
- The system shall display all user-facing text in Spanish.
- The system shall present relief information in plain, non-technical language.

Legal Requirements

- The system shall comply with the accessibility requirements of the Americans with Disabilities Act.
- The system shall comply with South Carolina's data breach notification law.
- The system shall obtain parental consent before collecting personal information from a user under the age of 13.

8. Definitions and Acronyms

The table below defines terms and acronyms used throughout this document that may not be immediately clear to a new reader.

Term / Acronym	Definition
SRS	Software Requirements Specification — this document.
Resident	A person affected by a hurricane who uses the platform to request help or find a shelter.
Volunteer	A person who registers to fulfill residents' requests for aid, transportation, or supplies.
Coordinator	The relief-organization staff member who verifies accounts, assigns volunteers, and broadcasts alerts.
Shelter Operator	The person responsible for registering a shelter and keeping its capacity and status current.
MoSCoW	A prioritization method — Must Have, Should Have, Could Have, Won't Have — used to rank requirements.
Need Request	A resident's submission describing a specific need, such as food, water, shelter, or medical supplies.
SCEMD	South Carolina Emergency Management Division.
WCAG	Web Content Accessibility Guidelines — the standard used to ensure the platform is usable by people with disabilities.
Uptime	The percentage of time the system is available and functioning correctly.

9. Competitive Analysis

	FEMA App
Strengths	Where the app really shines is the federal aid: It combines real-time NWS & IPAWS notifications in as many as five saved locations down to your specific zip code with a ready connection to disaster relief aid information, FAQs and local Disaster Recovery Centers. Providing a link into real federal assistance is one of the things that neither of the other two solutions do, and it comes with built in access for screen readers with built in access to the same federal disaster resources for disabled persons.
Weaknesses	A federal oversight report released last week confirms what has been feared about the app: survivors are still waiting for aid, struggling to reach caseworkers, and the aid-linking part of the app is limited by how inefficiently it can connect to bureaucracy. Real time, location aware tornado alerts are missing, which, given the number of hurricanes with spawned tornadoes, is a baffling oversight. It reads more like a piece of reactive bureaucracy than a personal, urgent crisis response.
Audience/Focus	The FEMA App is designed for all people preparing for, enduring, and recovering from any Federally declared disaster, regardless of hurricane involvement. The application is designated as the one, official, one-stop government application that allows for response to all phases of the disaster lifecycle from early warning up through application for recovery funds.

	Red Cross Emergency App
Strengths	Its “I’m Safe” function will notify any contact on your device if you’re ok and text a safety check message to friends who might not have the application themselves. The app will also provide around 40 customizable hazard notification settings, keep an eye on any five locations plus yourself and is able to override DND to send through to any of your “emergency” contacts. In addition, the app has several hurricane-ready features such as tracking, live weather maps, and a hurricane evacuation step-by-step guide.

Weaknesses	It does not have a way to receive funds like FEMA and with the short-term focus on shelter and safety checks, there is not a great use of the app for post- disaster recovery once those criteria are addressed. It has the capability of being an excellent resource management/communications tool, but it is not a recovery tool.
Audience/Focus	The Red Cross Emergency App speaks to homes, not government bureaucracy, and that can be hard to dismiss since they offer it in both English and Spanish. However, unlike the broad approach taken by FEMA that covers the entire money-application process, the Emergency App primarily focuses on the more straightforward need of getting everyone to a safe place and then maintaining communications.

	Crisis Cleanup
Strengths	The biggest win, however, is preventing duplicated effort by relief orgs helping in the same area. Support from the community's own efforts proves this, in a year the system facilitated survivors upload 73,000 photos of damage with a 96% user interaction rate – and an increasing number successfully closed or updated their own reports without human support. Even the hold music on the hotline, carefully developed with trauma-informed approaches in mind, reveals an understanding of human needs, not just a logistical plan.
Weaknesses	The obvious problem then is volunteer access: user-reviewers call the tool not usable, or indeed "simply, utterly useless," if the individual volunteer can't sign up, because it is organized around organization rather than the individual user as part of a volunteer. Users of the app are also concerned that data syncing doesn't work in a bad connection and, given that one of the places it's most-needed is in emergency situations (with few opportunities for good connection) that seem less-than-ideal.
Audience/Focus	Crisis Cleanup plays a different niche of the market all together. Instead of providing to individual storm survivors who want info and/or shelter it is a coordinating layer that brings together volunteer organizations and case managers on with the post storm debris removal, and the rebuild but it is on a secondary basis of survivors logging in requests and in order to utilize the service.

Summary

	Strengths	Weaknesses	Audience/Focus
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FEMA App	• Direct link to federal aid	• Slow, bureaucratic backend	• General public, federal aid
Red Cross Emergency App	• Family safety check-ins	• No financial aid path	• Families, shelter & safety
Crisis Cleanup	• Reduces duplicated relief efforts	• Hard for solo volunteers	• Volunteer orgs, cleanup coordination

After looking at the 3 products, there is 1 that stands out from the pack, the one that everyone in this area hasn't quite cracked: direct, unfettered matching of an individual unaffiliated volunteer with an individual disaster survivor who needs help directly with a small task, without first involving an organization on one side. FEMA's App is federal alerts and aid requests; it doesn't concern itself with coordinating volunteers at all. The Red Cross Emergency App concerns itself with shelters and safety check-ins, but a neighbor with a free pickup truck for the afternoon can't leverage it to coordinate with the folks down the road whose fence blew into their yard.

Crisis Cleanup does coordinate volunteers, but it's explicitly set up to facilitate organizational logins, as even Crisis Cleanup's own users have said it's 'utterly useless' if you're an individual trying to sign up on your own.

That is exactly where we want to come in. We aim to build a product for the person that doesn't currently affiliate with a non-profit (and doesn't want to) but wants to help out as soon as a storm has passed, as much as for the survivor who has a specific, small task that they don't want to have to wait on the phone line of a major organization to get done. FEMA will address the federal aid, Red Cross will address shelters and safety: we're taking over the lone area in which these services are lacking; immediate, intuitive, individualized matching of volunteers with survivors; no intermediate steps, no logins, nothing.

10. References

Sources referenced in the creation of this document:

- FEMA App — reviewed for the Competitive Analysis (Section 9).
- American Red Cross Emergency App — reviewed for the Competitive Analysis (Section 9).
- Crisis Cleanup — reviewed for the Competitive Analysis (Section 9).
- NBC News, “Hurricane Matthew Expected to Barrel Through Florida, Carolinas” — referenced for South Carolina evacuation statistics (Section 1, Introduction).
- Scientific American / U.S. Census Bureau, “Disasters Displaced More Than 3 Million Americans in 2022” — referenced for national disaster displacement statistics (Section 1, Introduction).

11. Appendices

The following supporting documents were produced alongside this requirements document and are available from the team:

- Appendix A: Client Discovery Questions — the list of questions prepared for the client requirements gathering session.
- Appendix B: Requirements & Notes Log — the spreadsheet template used by recorders during client meetings.
- Appendix C: Functional and Non-Functional Requirements Spreadsheet — the complete requirements list referenced in Sections 6 and 7.